

that are repeated just a handful of times. At the same time, we have shown that this memorization is often hard to discover, and for an attack to actually extract this data it will be necessary to develop qualitatively new attack strategies. Fortunately, it appears that (for the comparatively small models we study) training data inserted just once is rarely memorized, and so deduplicating training datasets (Lee et al., 2021) is likely a practical technique to mitigate the harms of memorization.

## REFERENCES

- Martin Abadi, Andy Chu, Ian Goodfellow, H Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. Deep learning with differential privacy. In *Proceedings of the 2016 ACM SIGSAC conference on computer and communications security*, pages 308–318, 2016.
- Rohan Anil, Badhi Ghazi, Vineet Gupta, Ravi Kumar, and Pasin Manurangsi. Large-scale differentially private BERT. *arXiv preprint arXiv:2108.01624*, 2021.
- Sid Black, Leo Gao, Phil Wang, Connor Leahy, and Stella Biderman. GPT-Neo: Large Scale Autoregressive Language Modeling with Mesh-Tensorflow, March 2021. URL <https://doi.org/10.5281/zenodo.5297715>. If you use this software, please cite it using these metadata.
- Hannah Brown, Katherine Lee, Fatemehsadat Mireshghallah, Reza Shokri, and Florian Tramèr. What does it mean for a language model to preserve privacy?, 2022.
- Nicholas Carlini, Chang Liu, Úlfar Erlingsson, Jernej Kos, and Dawn Song. The secret sharer: Evaluating and testing unintended memorization in neural networks. In *USENIX Security Symposium*, 2019.
- Nicholas Carlini, Florian Tramèr, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Úlfar Erlingsson, et al. Extracting training data from large language models. *arXiv preprint arXiv:2012.07805*, 2020.
- Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.
- Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. Calibrating noise to sensitivity in private data analysis. In *Theory of cryptography conference*, pages 265–284. Springer, 2006.
- William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *arXiv preprint arXiv:2101.03961*, 2021.
- Vitaly Feldman and Chiyuan Zhang. What neural networks memorize and why: Discovering the long tail via influence estimation. In *Advances in Neural Information Processing Systems*, 2020.
- Matt Fredrikson, Somesh Jha, and Thomas Ristenpart. Model inversion attacks that exploit confidence information and basic countermeasures. In *Proceedings of the 22nd ACM SIGSAC conference on computer and communications security*, pages 1322–1333, 2015.
- Karan Ganju, Qi Wang, Wei Yang, Carl A Gunter, and Nikita Borisov. Property inference attacks on fully connected neural networks using permutation invariant representations. In *Proceedings of the 2018 ACM SIGSAC conference on computer and communications security*, pages 619–633, 2018.
- Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, et al. The Pile: An 800GB dataset of diverse text for language modeling. *arXiv preprint arXiv:2101.00027*, 2020.
- Peter Henderson, Koustuv Sinha, Nicolas Angelard-Gontier, Nan Rosemary Ke, Genevieve Fried, Ryan Lowe, and Joelle Pineau. Ethical challenges in data-driven dialogue systems. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society*, pages 123–129, 2018.
- Matthew Jagielski, Jonathan Ullman, and Alina Oprea. Auditing differentially private machine learning: How private is private SGD? *arXiv preprint arXiv:2006.07709*, 2020.

- Bargav Jayaraman and David Evans. Evaluating differentially private machine learning in practice. In *28th {USENIX} Security Symposium ({USENIX} Security 19)*, pages 1895–1912, 2019.
- Nikhil Kandpal, Eric Wallace, and Colin Raffel. Deduplicating training data mitigates privacy risks in language models. *arXiv preprint arXiv:2202.06539*, 2022.
- Katherine Lee, Daphne Ippolito, Andrew Nystrom, Chiyuan Zhang, Douglas Eck, Chris Callison-Burch, and Nicholas Carlini. Deduplicating training data makes language models better. *CoRR*, abs/2107.06499, 2021. URL <https://arxiv.org/abs/2107.06499>.
- R. Thomas McCoy, Paul Smolensky, Tal Linzen, Jianfeng Gao, and Asli Celikyilmaz. How much do language models copy from their training data? Evaluating linguistic novelty in text generation using RAVEN. *CoRR*, abs/2111.09509, 2021. URL <https://arxiv.org/abs/2111.09509>.
- Milad Nasr, Shuang Song, Abhradeep Thakurta, Nicolas Papernot, and Nicholas Carlini. Adversary instantiation: Lower bounds for differentially private machine learning. *arXiv preprint arXiv:2101.04535*, 2021.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020. URL <http://jmlr.org/papers/v21/20-074.html>.
- Swaroop Ramaswamy, Om Thakkar, Rajiv Mathews, Galen Andrew, H Brendan McMahan, and Françoise Beaufays. Training production language models without memorizing user data. *arXiv preprint arXiv:2009.10031*, 2020.
- Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. Membership inference attacks against machine learning models. In *2017 IEEE Symposium on Security and Privacy (SP)*, pages 3–18. IEEE, 2017.
- Om Thakkar, Swaroop Ramaswamy, Rajiv Mathews, and Françoise Beaufays. Understanding unintended memorization in federated learning, 2020.
- Aleena Thomas, David Ifeoluwa Adelani, Ali Davody, Aditya Mogadala, and Dietrich Klakow. Investigating the impact of pre-trained word embeddings on memorization in neural networks. In *International Conference on Text, Speech, and Dialogue*, pages 273–281. Springer, 2020.
- Ben Wang and Aran Komatsuzaki. GPT-J-6B: A 6 Billion Parameter Autoregressive Language Model. <https://github.com/kingoflolz/mesh-transformer-jax>, May 2021.
- Samuel Yeom, Irene Giacomelli, Matt Fredrikson, and Somesh Jha. Privacy risk in machine learning: Analyzing the connection to overfitting. In *2018 IEEE 31st Computer Security Foundations Symposium (CSF)*, pages 268–282. IEEE, 2018.
- Chiyuan Zhang, Samy Bengio, Moritz Hardt, Benjamin Recht, and Oriol Vinyals. Understanding deep learning requires rethinking generalization. *ICLR*, 2017.
- Chiyuan Zhang, Daphne Ippolito, Katherine Lee, Matthew Jagielski, Florian Tramèr, and Nicholas Carlini. Counterfactual memorization in neural language models. *arXiv preprint arXiv:2112.12938*, 2021.
- Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, et al. Opt: Open pre-trained transformer language models. *arXiv preprint arXiv:2205.01068*, 2022.
- Albert Ziegler. GitHub Copilot: Parrot or crow? <https://docs.github.com/en/github/copilot/research-recitation>, 2021.

## A IMPLEMENTATION DETAILS FOR DATASET CREATION

Intuitively speaking, it is straightforward to construct a dataset containing specifiable proportions of documents at various frequencies. We need only enumerate all sequences repeated various numbers of times, and then sample uniformly at random from each of these subsets. However in practice this is difficult to do, given the scale of these datasets: even asking the question “how many times is this sequence present in the training dataset” requires linear work for each query, and so repeating this thousands of times for an 800GB dataset would be infeasible.

To do this efficiently, we build on the work of Lee et al. (2021) and construct a suffix array over the training dataset. Such a data structure allows efficient queries to enumerate all sequences of length  $k$  that are repeated between  $N$  and  $M$  times for any  $N, M$ . This can be accomplished by a linear scan of the suffix array. As notation, write  $i$  as the pointer into the dataset at a certain position  $j$  of the suffix array (i.e.,  $A[j] = i$ ),  $i'$  as the index at position  $j + N$  (so that  $A[j + N] = i'$ ), and  $i''$  as the index at position  $j + M$  (so that  $A[j + M] = i''$ ). Then, if  $D[i : i + k] = D[i' : i' + k]$  but  $D[i : i + k] \neq D[i'' : i'' + k]$ , the sequence  $D[k : i + k]$  is guaranteed to appear between  $N$  and  $M$  times in the dataset. As a result, we can scan linearly through the suffix array and enumerate all values of  $j$  to efficiently find all potential sequences repeated between  $N$  and  $M$  times. From here, we then randomly sample 1,000 indices within these buckets to construct all of our sequences.

## B LONGER DOCUMENTS ARE NOT EASIER TO MEMORIZE THAN SHORTER DOCUMENTS

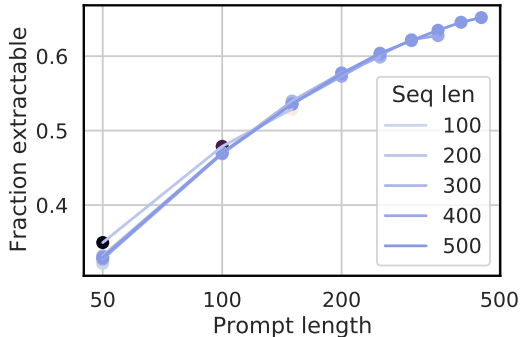


Figure 5: Longer sequences are not easier to extract. We compute the probability that an adversary can extract a sequence as a function of the number of tokens of context available, when varying the length of the sequences. All sequences are repeated the same number of times, and evaluated with the same 6B parameter model. Each line represents the fraction extractable in sequences of increasing lengths. Because all lines nearly perfectly overlap, longer sequences are not fundamentally “easier” to extract than shorter sequences.

Intuitively, one might think that longer sequences are more likely in the tail of the distribution, and if the model is trained to a low perplexity, then the tail of the distribution may be more likely to be memorized. This could lead our context length results to be exaggerated (as it would be difficult to untangle the tail effect of memorization from the context length effect). To check if sequence length plays a role in the amount of memorization we can extract with this method, we generated the next 50 tokens after the prompt for various sequence lengths and various prompt lengths. Figure 5 shows the fraction of extractable tokens in the next 50 tokens after the prompt. Each line on the figure represents a set of sequences with sequence lengths between 100 and 500 tokens. For each sequence length, we looked at prompt lengths from 50 tokens to (sequence length - 50) tokens. We do not see significant differences between the fraction of extractable tokens with varying prompt lengths across various sequence lengths.